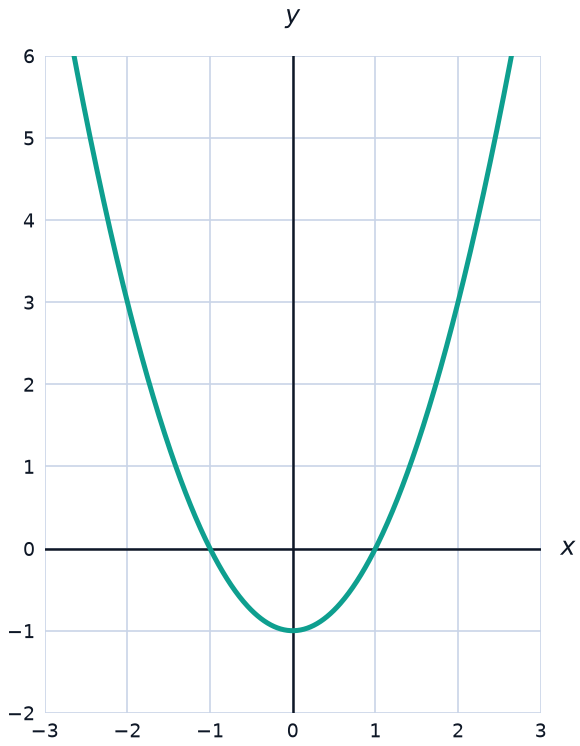
Parametric Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Eliminating the parameter**

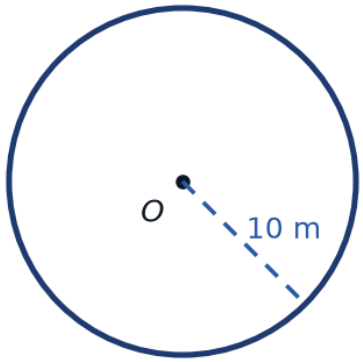
Eliminate the parameter from $x(t) = t$, $y(t) = t^2 - 1$ to find the Cartesian equation, and sketch it.



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2 [Foundational] • Target: Parametric equations from real motion

A Ferris wheel of radius 10 m completes one revolution every 40 s, centered 12 m above the ground. Write parametric equations for a rider's height above the ground as a function of time t (seconds), assuming the rider starts at the bottom.



3 [Foundational] • Target: Derivatives of parametric equations

For $x(t) = t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ in terms of t.

4 [Foundational] • Target: Derivatives of parametric equations

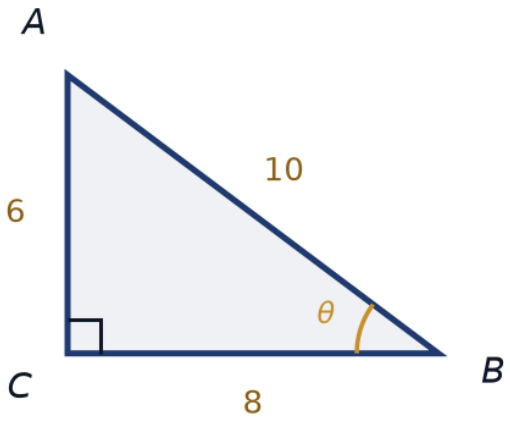
For $x(t) = \cos t$, $y(t) = \sin t$, find $\frac{dy}{dx}$ and evaluate it at $t = \frac{\pi}{4}$.

5 [Foundational] • Target: Second derivatives of parametric equations

For $x(t) = t^2$, $y(t) = t^3$, find $\frac{d^2y}{dx^2}$ using $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$.

6 [Foundational] • Target: Tangent lines to parametric curves

Find the equation of the tangent line to the curve $x(t) = t^2 - 1$, $y(t) = t^3 - t$ at $t = 2$.



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7 [Foundational] • Target: Arc length of parametric curves

State the arc length formula for a parametric curve $x(t)$, $y(t)$ on $[a, b]$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Arc length of parametric curves

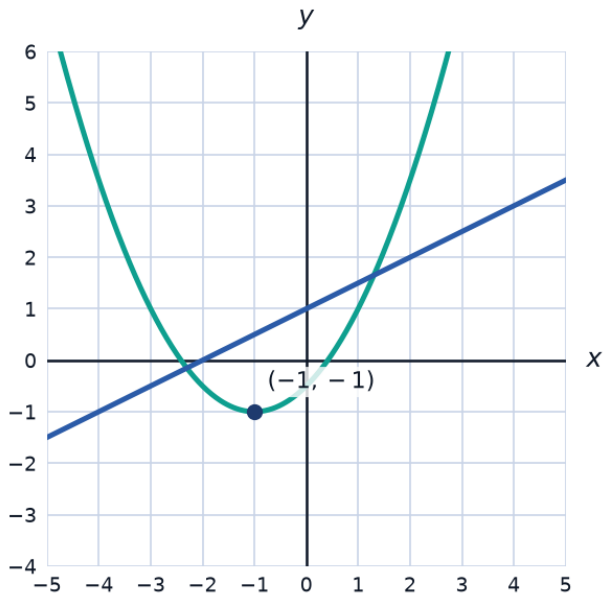
Find the arc length of $x(t) = 3t$, $y(t) = 4t$ for $t \in [0, 2]$.

9 [Intermediate] • Target: Arc length of parametric curves

Set up (but do not evaluate) the arc length integral for $x(t) = t^2$, $y(t) = t^3$ on $[0, 1]$.

10 [Intermediate] • Target: Speed as a vector quantity

A particle moves with position $x(t) = t^2$, $y(t) = 2t$. Find its speed at $t = 1$.



11 [Intermediate] • Target: Speed as a vector quantity

A particle moves with position $x(t) = 3\cos t$, $y(t) = 3\sin t$. Find its speed at any time t, and interpret the result.

12 [Intermediate] • Target: Eliminating the parameter

Eliminate the parameter from $x(t) = 2t + 1$, $y(t) = 4t - 3$ to find the Cartesian equation.

13 [Intermediate] • Target: Eliminating the parameter

Explain why eliminating the parameter from $x(t) = \cos t$, $y(t) = \sin t$ using the identity $\sin^2 t + \cos^2 t = 1$ gives a relation that is NOT a function $y = f(x)$.

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14 [Intermediate] • Target: Concavity of parametric curves

For $x(t) = t^2$, $y(t) = t^3$ (with $\frac{d^2y}{dx^2} = \frac{3}{4t}$ found earlier), determine the concavity of the curve for $t > 0$ and for $t < 0$.

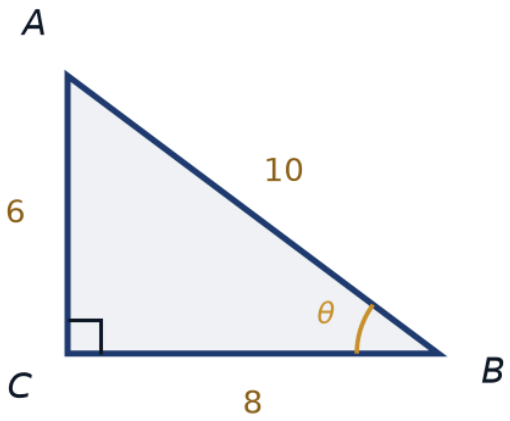
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Horizontal and vertical tangents • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

For $x(t) = t^2 - 4$, $y(t) = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal (where $\frac{dy}{dt} = 0$, provided $\frac{dx}{dt} \neq 0$).

16 [Challenging] • Target: Horizontal and vertical tangents • Cross-Lesson: 6.01 Power Series and Interval of Convergence

For the same curve, find the value(s) of t where the tangent line is vertical (where $\frac{dx}{dt} = 0$, provided $\frac{dy}{dt} \neq 0$).



17 [Challenging] • Target: Motion along a parametric path (particle interpretation) • Cross-Lesson: 6.02 Taylor and Maclaurin Series

A particle's position is $x(t) = t^2$, $y(t) = t^3 - 3t$ for $t \in [-2, 2]$. Find all times when the particle is moving straight up or down (vertical velocity component only, i.e. $\frac{dx}{dt} = 0$).

18 [Challenging] • Target: Motion along a parametric path (particle interpretation) • Cross-Lesson: 1.01 Integration by Parts

For the particle above, find the speed at $t = 2$.

19 [Challenging] • Target: Arc length of a full circle (verifying the circumference formula) • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Use the parametric arc length formula to verify that the circumference of a circle $x(t) = r\cos t$, $y(t) = r\sin t$ for $t \in [0, 2\pi]$ is $2\pi r$.

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20 [Challenging] • Target: Arc length of a full circle (verifying the circumference formula) • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Find the arc length of the parametric curve $x(t) = e^t \cos t$, $y(t) = e^t \sin t$ is a spiral; set up (but do not evaluate) its arc length integral on $[0, \pi]$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Derivatives of parametric equations

For $x(t) = 6t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 3$.

a. $-\frac{3}{2}$

b. -6

c. $\frac{3}{2}$

d. $-\frac{1}{2}$

22 [Foundational] • Target: Second derivatives of parametric equations

For $x(t) = 2t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 6$.

a. -4

b. 9

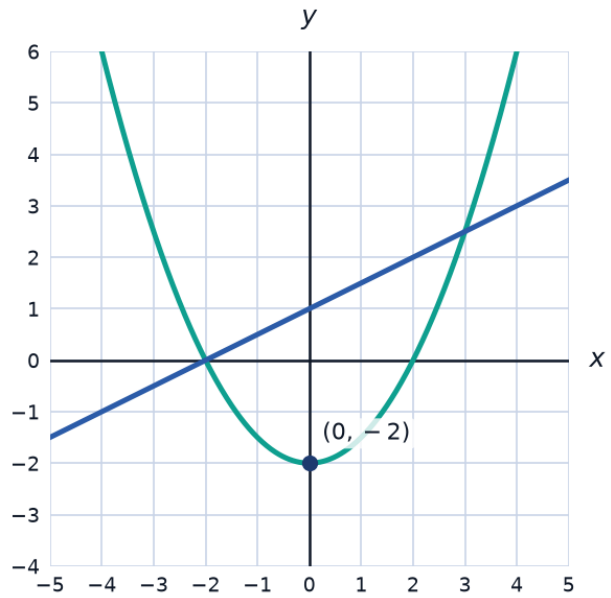
c. -9

d. -8

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23 [Foundational] • Target: Tangent lines to parametric curves

For $x(t) = 4t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 3$.



a. $\frac{9}{4}$

b. -4

c. $-\frac{9}{4}$

d. $-\frac{5}{4}$

24 [Foundational] • Target: Arc length of parametric curves

For $x(t) = 4t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ at $t = 4$.

a. $\frac{5}{2}$

b. $\frac{3}{2}$

c. $\frac{32}{3}$

d. $-\frac{3}{2}$

25 [Foundational] • Target: Speed as a vector quantity

For $x(t) = 6t^2$, $y(t) = -3t^3$, find $\frac{dy}{dx}$ at $t = 1$.

a. $-\frac{4}{3}$

b. $-\frac{3}{4}$

c. $\frac{1}{4}$

d. $\frac{3}{4}$

26 [Foundational] • Target: Eliminating the parameter

For $x(t) = t^2$, $y(t) = 5t^3$, find $\frac{dy}{dx}$ at $t = 4$.

a. $\frac{8}{15}$

b. -30

c. 31

d. 30

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27 [Foundational] • Target: Concavity of parametric curves

For $x(t) = 6t^2$, $y(t) = 2t^3$, find $\frac{dy}{dx}$ at $t = 5$.

a. $-\frac{5}{2}$

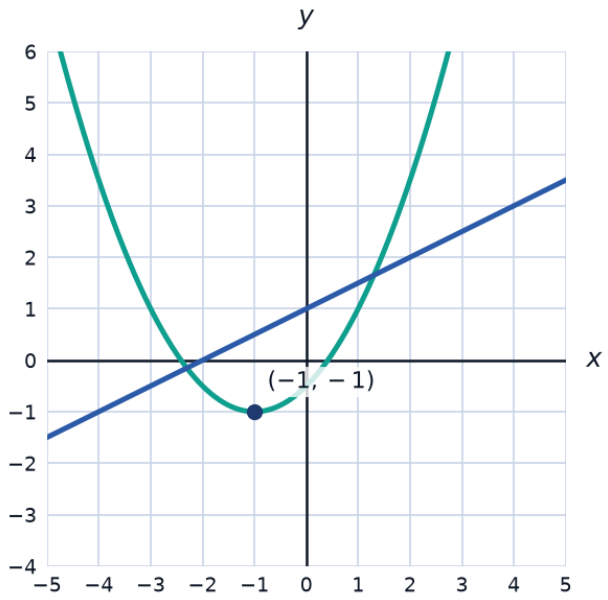
b. $\frac{7}{2}$

c. 10

d. $\frac{5}{2}$

28 [Foundational] • Target: Horizontal and vertical tangents

Eliminate the parameter from $x(t) = -2t - 3$, $y(t) = 7t - 9$ to find the Cartesian equation, then state its slope.



a. $-\frac{7}{2}$

b. $-\frac{5}{2}$

c. $-\frac{2}{7}$

d. $\frac{7}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Motion along a parametric path (particle interpretation)

Eliminate the parameter from $x(t) = 2t - 9$, $y(t) = -t + 4$ to find the Cartesian equation, then state its slope.

a. $-\frac{1}{2}$

b. -2

c. $\frac{1}{2}$

d. 0

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30 **[Intermediate]** • **Target: Arc length of a full circle (verifying the circumference formula)**

Eliminate the parameter from $x(t) = -5t - 4$, $y(t) = -4t - 7$ to find the Cartesian equation, then state its slope.

a. $-\frac{4}{5}$

b. $\frac{5}{4}$

c. $\frac{4}{5}$

d. $\frac{9}{5}$

31 **[Intermediate]** • **Target: Parametric equations from real motion**

Eliminate the parameter from $x(t) = t + 1$, $y(t) = 2t + 5$ to find the Cartesian equation, then state its slope.

a. 3

b. -2

c. $\frac{1}{2}$

d. 2

32 **[Intermediate]** • **Target: First and Second Derivatives Parametrically**

Eliminate the parameter from $x(t) = -4t + 3$, $y(t) = -t + 5$ to find the Cartesian equation, then state its slope.

a. $-\frac{1}{4}$

b. $\frac{1}{4}$

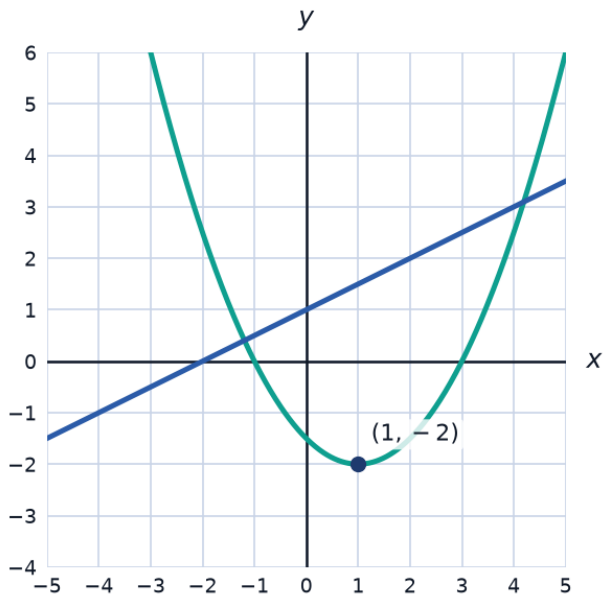
c. 4

d. $\frac{5}{4}$

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33 [Intermediate] • Target: Horizontal and Vertical Tangents

Eliminate the parameter from $x(t) = 4t + 1$, $y(t) = 9t + 7$ to find the Cartesian equation, then state its slope.



a. $-\frac{9}{4}$

b. $\frac{13}{4}$

c. $\frac{9}{4}$

d. $\frac{4}{9}$

34 [Intermediate] • Target: Arc Length of Parametric Curves

Eliminate the parameter from $x(t) = 3t - 1$, $y(t) = 2t + 2$ to find the Cartesian equation, then state its slope.

a. $-\frac{2}{3}$

b. $\frac{2}{3}$

c. $\frac{5}{3}$

d. $\frac{3}{2}$

35 [Intermediate] • Target: Speed and Total Distance Traveled

For $x(t) = 3t^2 + 5$, $y(t) = t^3 - 4t$, find $\frac{dy}{dx}$ at $t = 1$.

a. $\frac{1}{6}$

b. $\frac{5}{6}$

c. -6

d. $-\frac{1}{6}$

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36 [Intermediate] • Target: Derivatives of parametric equations

For $x(t) = 3t^2 - 5$, $y(t) = 5t^3 + 3t$, find $\frac{dy}{dx}$ at $t = 3$.

a. $\frac{26}{3}$

b. $\frac{3}{23}$

c. $-\frac{23}{3}$

d. $\frac{23}{3}$

37 [Intermediate] • Target: Second derivatives of parametric equations

For $x(t) = t^2$, $y(t) = t^3 - 9t^2 + 24t$, find the larger value of t where the tangent line is horizontal.

a. 2

b. 4

c. 6

d. 8

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Tangent lines to parametric curves • Cross-Lesson: 1.01 Integration by Parts

For $x(t) = t^2 - 3t$, $y(t) = t^3 + 3t^2 - 24t$, find the larger value of t where the tangent line is horizontal.

a. -4

b. -2

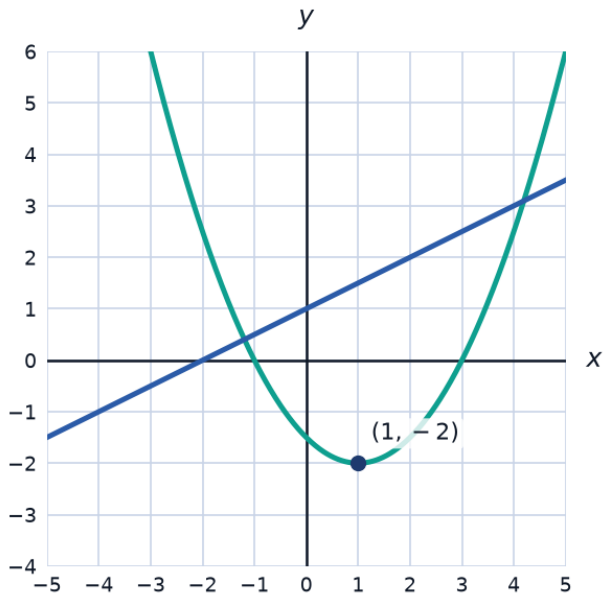
c. -8

d. 2

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39 [Challenging] • Target: Arc length of parametric curves • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

A particle moves with position $x(t) = t^2$, $y(t) = 6t$. Find its speed at $t = 4$.



a. 12

b. 8

c. 6

d. 10

40 [Challenging] • Target: Speed as a vector quantity • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

A particle moves with position $x(t) = t^2$, $y(t) = 8t$. Find its speed at $t = 3$.

a. 6

b. 10

c. 8

d. 12

41 [Challenging] • Target: First and Second Derivatives Parametrically • Cross-Lesson: 3.02 Vector-Valued Functions

A curve is defined by $x(t) = t^2 + 1$ and $y(t) = t^3 - 3t$. Find $\frac{dy}{dx}$ at $t = 2$.

a. $\frac{9}{4}$

b. $\frac{3}{2}$

c. $\frac{4}{9}$

d. 9

42 [Challenging] • Target: First and Second Derivatives Parametrically • Cross-Lesson: 4.01 Polar Curves and Derivatives

For $x(t) = t^2$ and $y(t) = t^3$, find $\frac{d^2y}{dx^2}$.

a. $\frac{3}{4t}$

b. $\frac{3}{2}$

c. $\frac{6t}{2}$

d. $\frac{3}{4t^2}$

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43 [Challenging] • Target: Horizontal and Vertical Tangents • Cross-Lesson: 4.02 Area in Polar Coordinates

Find the equation of the tangent line at $t = 1$ to the curve $x(t) = e^{2t}, y(t) = 3t^2 - 1$.

a. $y - 2 = \frac{3}{e^2}(x - e^2)$

b. $y - 2 = 3(x - e^2)$

c. $y = \frac{6}{e^2}x$

d. $y - 1 = \frac{3}{e^2}(x - e^2)$

44 [Challenging] • Target: Arc Length of Parametric Curves • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Find the arc length of the curve $x = 3\cos t, y = 3\sin t$ from $t = 0$ to $t = \pi$.

a. 3π

b. 6π

c. 9π

d. $\frac{3\pi}{2}$

45 [Challenging] • Target: Speed and Total Distance Traveled • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

A particle moves in the xy-plane with velocity vector $\mathbf{v}(t) = \langle 4t, 3 \rangle$. What is its speed at $t = 1$?

a. 5

b. 7

c. 4

d. $\sqrt{7}$

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